

EVOLUTION Series

PD25

IF Satellite Modem



OVERVIEW

The Evolution Series PD25 has been designed for cost-critical modem applications and discerning users who demand quality and reliability at an affordable price. This **25Mbps** capable modem offers full compliance with IESS-308, 309, 310, 314 & 315, plus a range of data interfaces including Ethernet. The Evolution Series Satellite Modem design is based on highly programmable logic giving the flexibility of instant feature upgrades and built-in future-proofing.

Advanced Bandwidth-Efficient Features

Evolution Series Modems contain a host of bandwidth-efficient features, which can all be used at the same time.

Paired Carrier™ overlays transmit and receive carriers reducing satellite bandwidth by up to 50%. Paired Carrier™ uses ViaSat's patented PCMA technology.

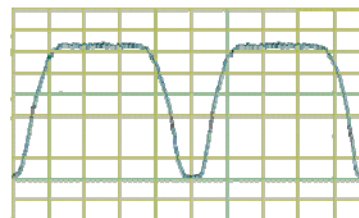
FastLink™ low-latency LDPC has been designed specifically for latency-sensitive applications while giving coding gain that is close to the theoretical limits.

Advanced bandwidth-saving IP features include acceleration and header and payload compression. A sophisticated on-board IP traffic shaping feature allows end-to-end provisioning of quality of service.

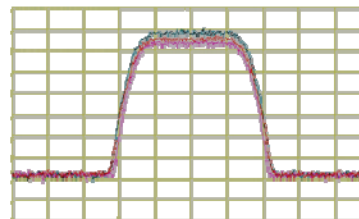
FEATURES

- ▶ Data rate options to 25Mbps, 12.5Mps.
- ▶ Paired Carrier™ option.
- ▶ A wide range of terrestrial interfaces including Ethernet, serial and G.703.
- ▶ Advanced IP feature set including TCP acceleration, compression, routing, bridging, traffic shaping, ACM, VCM and throughput/diagnostic graphs.
- ▶ FastLink Low-Latency LDPC, 2nd Generation Turbo (TPC) and other FEC options.
- ▶ Modulations up to 64QAM.
- ▶ **New!** Patent-pending LinkGuard™ signal-under-carrier interference detection.

Paired Carrier™ Operation



Paired Carrier
Disabled



Paired Carrier
Enabled
Can save 50%
on space segment

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Main Specifications	
Frequency	50 to 90MHz & 100 to 180MHz (resolution 100Hz) (BNC connector)
Data Rate	4.8kbps to 25Mbps 1bps resolution Operation to 2,048kbps provided as standard; extension options to 5Mbps, 10Mbps, 25Mbps
Symbol Rate	9.6ksps to 12.5Msps
Operating Modes	Closed Network (+ESC) (IESS-315) IBS/IDR (IESS-308/309/310/314) options
Scrambling	IBS: Synchronised to framing per IESS-309 IDR with RS coding: Synchronised to RS overhead IDR, no RS coding, non-TPC FEC: V.35 self-synchronising IDR, no RS coding, with TPC FEC: 2 ⁿ 12-1 up to 10 Mbps Closed+ESC: Synchronised to ESC overhead
IF Impedance	50Ω & 75Ω
Return Loss	18dB typical
External Reference	Clocking only: 1 to 10MHz, 1kHz steps Clocking and RF frequency: 10MHz, 0dBm±1dB
Redundancy	Can be operated in standalone, 1:1 or 1:N redundancy configuration

Traffic Interfaces	
Base modem (standard): Ethernet (10/100 BaseT) IP traffic on RJ45	
Traffic options: IP Traffic card 10/100/1000 BaseT on RJ45 (increases performance compared to base modem IP traffic) RS422, X.21, V.35 and RS232 on EIA530 connector (25-pin D-type female) Serial LVDS (25-pin D-type female) G.703 (balanced on EIA530) G.703 (unbalanced on BNC 75Ω female) Quad E1 G.703 (balanced on RJ45) HSSI (50-pin HD SCSI-2 connector)	
MultiMux option: generates a single carrier from any mixture of G.703, IP and EIA530 traffic	

Modulator	
Output Power	0 to -25dBm (0.1dB steps)
Output Power Stability	±0.5dB, 0°C to 50°C
Transmit Filter Roll-off	20%, 25%, 35%
Phase Accuracy	±2° maximum
Amplitude Accuracy	±0.2dB maximum
Carrier Suppression	-30dBc minimum
Output Phase Noise	As IESS-308, nominally 3dB better
Frequency Stability	<1ppm/yr
Harmonics	Better than -55dBc/ 4kHz in band
Spurious	Better than -55dBc/ 4kHz in band
Transmit On/Off Ratio	55dB minimum
Adaptive Signal Predistorter Option	Use with 16QAM to relax HPA backoff by up to 1.6dB. Compensates for HPA non-linearities

Demodulator	
Input Range	-30 to -60dBm wanted signal
Maximum Composite Signal	30dB above level of desired input up to a maximum of 0dBm
Frequency Sweep Width	±1kHz to ±32kHz up to 10 Msps (1kHz steps) ±10kHz to ±250kHz above 10 Msps (10kHz steps)
Acquisition Threshold	<5dB Es/No QPSK
Acquisition Time	Dependent on FEC, data rate and sweep width (at 9.6kbps, less than 1s at 6dB Es/No QPSK; at 10Mbps, less than 100ms at 6dB Es/No QPSK)
Clock Tracking Range	±100ppm minimum
Receive Filter Roll-off	20%, 25%, 35%
Performance Monitoring	Eb/No (range 0-15dB, ±0.2dB) Frequency offset (100Hz resolution) Receive signal level Buffer fill status
AGC Output	Buffered direct AGC output for antenna tracking, etc.

Forward Error Correction	
Modulation	BPSK, QPSK, OQPSK plus options for: 8PSK, 16QAM, FastLink 8QAM, FastLink 16APSK, FastLink 32APSK, FastLink 64QAM
FEC	Note BPSK and (O)QPSK provided as standard; other modulations are options FastLink Low-Latency LDPC option: BPSK 0.499 (O)QPSK 0.532, 0.639, 0.710, 0.798 8PSK/8QAM: 0.639, 0.710, 0.778 16APSK/16QAM: 0.726, 0.778, 0.828, 0.851 32APSK: 0.778, 0.828, 0.886, 0.938 64QAM: 0.828, 0.886, 0.938, 0.960 TPC option: BPSK 5/16, 21/44, 0.493 (Paradise), 2/3, 3/4, 0.789 (Paradise), 7/8 (Paradise), Rate 7/8 de facto (O)QPSK: 5/16, 21/44, 0.493 (Paradise), 2/3, 3/4, 0.789 (Paradise), 7/8 (Paradise), 7/8 de facto, 0.93 (Paradise) 8PSK: 3/4 de facto, 7/8 de facto, 0.93 (Paradise) 16QAM: 3/4 de facto, 7/8 de facto, 0.93 (Paradise) Viterbi: BPSK/(O)QPSK 1/2, 3/4, 7/8 TCM option: 8PSK 2/3 Sequential option: BPSK/(O)QPSK 1/2, 3/4, 7/8 Reed-Solomon outer codec available with Viterbi and TCM

Ethernet Traffic	
Throughput Performance	The maximum modem throughput depends on IP traffic format and the features enabled. Bridged IP/UDP data can be processed up to the modem maximum data rate. Please seek assistance from Paradise Datacom in evaluating your particular requirements.
Routing and Bridging	Bridging (standard). Static routing (standard). Dynamic routing option: RIP V1, V2; OSPF V2, V3; BGP V4
TCP Acceleration Option	Typical throughput level of 90% of link capacity. IP Traffic card option: Supports 5,000 concurrent accelerated TCP connections (plus at least 35,000 unaccelerated TCP connections) up to the modem maximum data rate. Base modem TCP acceleration option is restricted to 1000 accelerated TCP connections and 10Mbps. IP Traffic card includes HTTP Acceleration (reduces web page download times)
Header Compression Option	IP Traffic card option. Robust Header Compression to RFC 3095. Reduces Ethernet/IP/UDP/RTP header sizes typically by 90%. 1-way packet processing limit: 29,000 pps; 2-way limit: 22,000 pps. Includes Ethernet header compression (compresses 14-byte Ethernet frame to typically one byte)
Traffic Shaping Option	Provides guaranteed throughput levels for IP streams, using Committed Information Rate and Burst Information Rate settings. Stream differentiation is by IP address, IEEE 802.1p priority class, Diffserv DSCP class or MPLS EXP field
Encryption Option	Encrypts all IP traffic using AES with 256-bit keys
VLAN Support	IEEE 802.1q VLAN support (standard) IEEE 802.1p Quality of Service (packet prioritisation) using strict priority or fair weighting queuing
DHCP, SNMP	DHCP (standard) for automatic allocation of M&C IP address. SNMP (standard) v1, v2c and v3
Web Server	Embedded web server M&C interface (standard)
IP Diagnostic Graphs	Shows Tx, Rx throughput (bps, pps); dropped, errored packet counts (standard)

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Paired Carrier

Paired Carrier	Transmit and receive carriers are overlaid on top of each other in the same space segment. Echo cancellation techniques are used in the demodulator to cancel the transmit carrier and extract the wanted receive carrier signal
Paired Carrier data rate options	256kbps, 512kbps, 1024kbps, 2.5Mbps, 5Mbps, 10Mbps, 15Mbps, 20Mbps, 25Mbps (30kHz minimum occupied bandwidth; operates to maximum symbol rate of modem)
Supported power asymmetry	-10dB to +10dB
Supported symbol rate asymmetry	Up to 12:1
Eb/No degradation	Typically < 0.5dB (0.7dB for 16QAM/16APSK with 10dB power asymmetry)
Mobile Operation	Uses GPS data to continually recalculate position relative to satellite, allowing uninterrupted operation in mobile environments (ships, etc.) anywhere in satellite footprint

Drop & Insert Option

Bearer Types	T1-D4, T1-ESF, E1-G.732
Timeslot Selection	Independent selection of arbitrary timeslots for both drop and insert.
Bearer Generation	Terrestrial bearer may be looped through modem, or terminated after Drop Mux and a new bearer generated by the insert Mux
Timeslot ID	Maintains the identity of individual Drop/Insert timeslots for N=1,2,3,4,5,6,8,10,12,15,16, 20, 24 and 30. (See extended option below)

Extended Drop & Insert Option

Timeslot Re-Ordering	Selected timeslots may be independently re-ordered on both Tx and Rx paths
Multi-Destinational Working	All or only a subset of the received data may be inserted into the terrestrial bearer on the receive path for multi-destinational working
Timeslot ID Maintenance	The framed service is extended to maintain the identity of individual timeslots for all values of N from 1 to 31
Signaling	CAS and RBS are fully supported

Advanced ESC

ESC/Aux Port	Provides high rate async ESC or Intelsat low rate async IBS ESC	
Electrical Interface	IP, RS232, RS422 or RS485	
Async ESC	Closed Net Plus ESC	Overhead scales to any ESC baud rate from 0.5% to 70% of the main channel rate
	IBS Option	High rate async channel (1/32nd to 2/32nd of the IBS overhead) providing async baud rates from 0.2% to 5.1% of the terrestrial rate
Advanced Aux	Intelsat low-rate async ESC carried in bit 1 of TS32 providing a synchronous channel at 1/480th of the data rate, allowing up to one quarter of this rate for over-sampled async data	

Guaranteed Eb/No BER Performance (dB) (Typical in brackets)

		Rate 1/2	Rate 3/4	Rate 7/8	Rate 2/3	Rate 0.93
Viterbi QPSK	1E-4	4.7 (4.4)	6.1 (5.8)	7.1 (6.8)		
	1E-8	7.2 (6.9)	8.8 (8.5)	9.5 (9.2)		
Sequential (64kbps)	1E-4	4.3 (4.0)	5.4 (5.1)	6.4 (6.1)		
	1E-8	6.4 (6.1)	7.3 (7.0)	8.6 (8.3)		
Sequential (2048kbps)	1E-4	5.6 (5.3)	6.1 (5.8)	6.9 (6.6)		
	1E-8	7.5 (7.2)	8.1 (7.8)	8.4 (8.1)		
Turbo (TPC) QPSK	1E-4	2.7 (2.4)	3.5 (3.2)	4.1 (3.8)		
	1E-6					6.3 (6.0)
	1E-8	3.3 (3.0)	4.5 (4.2)	4.5 (4.2)		6.8 (6.5)
Turbo (TPC) 8PSK	1E-4		5.6 (5.3)	6.8 (6.5)		
	1E-6					9.2 (8.9)
	1E-8		6.8 (6.3)	7.2 (6.8)		9.9 (9.6)
Turbo (TPC) 16QAM	1E-3		6.5 (6.2)	7.7 (7.4)		
	1E-6					10.0 (9.7)
	1E-7		7.8 (7.5)	8.2 (7.8)		
	1E-8					10.7 (10.4)
8PSK/TCM	1E-3				6.3 (6.0)	
	1E-8				10.4 (10.1)	
8PSK/TCM + Reed-Solomon (all rates)	1E-4				6.1 (5.8)	
	1E-10				7.3 (7.0)	

FASTLINK LOW-LATENCY LDPC: SEE SEPARATE DATASHEET

EZ BERT Option

BER Channel	Bit error rate tester operates over main traffic, ESC or Aux channels, allowing BER monitoring while on traffic
Test Patterns	Various test patterns compatible with common BERTesters
Other test modes	Transmit CW (pure carrier) Transmit alternate 1-0 pattern Simulated satellite delay for TCP/IP packets

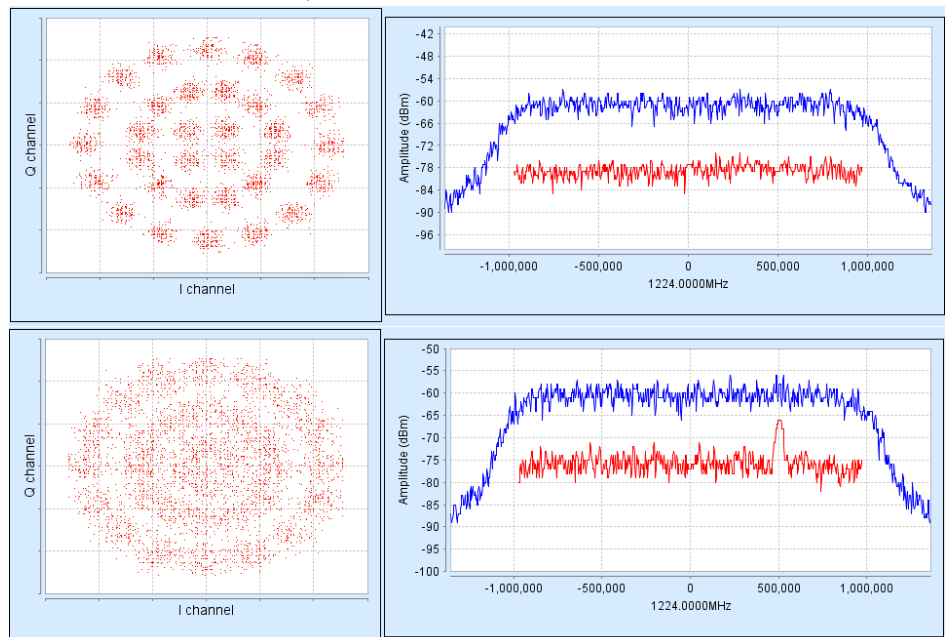
ODU facilities via IF interface

FSK Control Option	Allows monitor & control of a compatible Transceiver from the modem via the Tx IFL cable
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Mechanical/Environmental

Size	1U chassis, 410mm deep excluding front panel handles and rear panel connectors and fans
Weight	3.5kg
Power Supply	100-240VAC, +6%, -10%, 1A @100V, 0.5A @ 240V, 47-63Hz Fused IEC connector (live and neutral fused); 48V DC option
Safety Standards	EN60950-1
Emission and Immunity	EN55022 Class B (Emissions) EN55024 (Immunity)
Operating Temperature	0 to 50°C
Humidity	95% relative humidity, non-condensing
Compliance	FCC, CE and RoHS compliant
Alarm Relays	4 Independent Form C relays for unit, Tx, Rx and backward alarms

Built-in Spectrum Analyser showing **LinkGuard™** Signal-Under-Carrier interference detection without/with interferer present.



Fully configurable - pay only for what you need!

	Option	Description
Base Modem	✓	4.8kbps to 2.048Mbps closed network modem with two Ethernet 10/100 BaseT RJ45s for M&C and traffic respectively; Ethernet bridge, static routing; IPv4/IPv6 support IF 70MHz & 140MHz operation (50 to 90MHz and 100 to 180MHz) BPSK/QPSK/OQPSK; Viterbi FEC rates 1/2, 3/4 & 7/8; Intelsat Reed-Solomon outer codec Advanced ESC: Variable rate Async channel for Closed Net plus ESC operation AUPC: Automatic Uplink Power Control Web browser monitoring tools: Spectrum Display, Constellation Monitor, TCP/IP throughput IEEE 802.1p QoS; IEEE 802.1q VLAN support
Data Rate Options		5Mbps data rate: extends base operation to 5Mbps
		10Mbps data rate: extends 5Mbps operation to 10Mbps
		20Mbps data rate: extends 10Mbps operation to 25Mbps
IP Options <i>(all features require IP Traffic card other than 10Mbps TCP acceleration)</i>		Traffic Shaping: supports CIR/BIR/priority settings for IP streams classified by IP address, Diffserv class, IEEE 802.1p priority tag or MPLS EXP field
		Header Compression: IP/UDP/TCP/RTP packet header compression (RFC 3095) plus Ethernet header compression
		Payload Compression: TCP/UDP packet payload compression using the Deflate algorithm (RFC 1951)
		Encryption: TCP/IP packet payload encryption using AES with 256-bit keys
		Dynamic Routing: RIP, OSPF, BGP plus static routes
		Web Page Acceleration: acceleration of HTTP requests through pre-fetching of web page contents (requires TCP Acceleration)
		TCP Acceleration: to 10Mbps, subject to prevailing modem data rate limits
		TCP Acceleration: extends 10Mbps operation to 25Mbps, subject to prevailing modem data rate limits
Position 1 (must choose 1 option) hardware option		EIA-530 (D25 DCE providing selectable RS422/X.21/V.35/RS232, also balanced G.703 if G.703 option fitted)
		IDR (IESS 308)
		Blank panel
Position 2 (must choose 1 option) hardware option		IP Traffic card (2x10/100/1000 BaseT RJ45)
		EIA-530 (D25 DCE providing RS422/X.21/V.35/RS232, also balanced G.703 if G.703 option fitted)
		Quad E1 Multiplexer (balanced G.703 on 4xRJ45 of which one is enabled by default; includes Drop & Insert and IBS satellite framing)
		Serial LVDS (on D25)
		HSSI (on HD50 50-way SCSI-2 connector)
		Blank panel
Position 2 Quad E1 Mux options (only used with Quad E1 Mux card)		Adds Port 2 with Drop & Insert (requires Quad E1 Mux plus data rate option to 5Mbps)
		Adds Port 3 with Drop & Insert (requires Quad E1 Mux with Port 2 option plus data rate options to 10Mbps)
		Adds Port 4 with Drop & Insert (requires Quad E1 Mux with Port 2 & 3 options plus data rate options to 10Mbps)
		MultiMux: multiplexes any mixture of E1, IP and EIA-530 traffic types onto a single carrier; see separate Quad E1 application note for further details
Position 3 (must choose 1 option) hardware option		No BNC traffic interface
		2 x BNC sockets (unbalanced G.703 75Ω - supplied only with G.703 option)
Low-rate TPC <i>Subject to prevailing data rate limits</i>		Rates 5/16, 21/44, 3/4 in BPSK, QPSK, OQPSK; Rate 7/8 in QPSK, OQPSK; Rate 0.93 Paradise in QPSK, OQPSK; Rates 3/4, 7/8, 0.93 Paradise in 8PSK (requires 8PSK option); Rates 3/4, 7/8, 0.93 Paradise in 16QAM (requires 16QAM option) (10Mbps maximum data rate)
High-rate TPC <i>Extension to 25Mbps subject to prevailing data rate limits</i>		Rates 5/16, 21/44, 3/4 in BPSK, QPSK, OQPSK; Rate 7/8 in QPSK, OQPSK; Rate 0.93 Paradise in QPSK, OQPSK; Rates 3/4, 7/8, 0.93 Paradise in 8PSK (requires 8PSK option); Rates 3/4, 7/8, 0.93 Paradise in 16QAM (requires 16QAM option) (Requires Low-rate TPC option)
LinkGuard™		Signal-under-carrier interference detection web spectrum graph showing received spectrum and any interference underneath the received carrier while on traffic; automated alarm when interference rises above user-set threshold; supported for all non-DVB-S2 FECs and modulations
G.703		E1, T1, E2, T2 interfaces (hardware option) - requires either EIA-530 in Position 1 or 2 or BNC sockets fitted in Position 3

Configuration options continue on next page.

Fully configurable - pay only for what you need!

Option	Description
Paired Carrier™ <i>Subject to prevailing modem data rate limits. Occupied bandwidth: minimum 30kHz; operates to maximum symbol rate of modem</i>	Paired Carrier™ hardware option (requires one or more options below); allows Tx & Rx carriers to be overlapped, reducing the required satellite bandwidth
	Paired Carrier™ up to 256kbps (requires Paired Carrier™ hardware option)
	Extends Paired Carrier™ up to 512kbps
	Extends Paired Carrier™ up to 1.024Mbps
	Extends Paired Carrier™ up to 2.5Mbps
	Extends Paired Carrier™ up to 5Mbps
	Extends Paired Carrier™ up to 10Mbps
	Extends Paired Carrier™ up to 15Mbps
	Extends Paired Carrier™ up to 20Mbps
	Extends Paired Carrier™ up to 25Mbps
FastLink™ Low-latency LDPC FEC <i>subject to prevailing modem data rate limits</i>	FastLink™ LDPC hardware option (requires one or more additional FastLink™ options below); BPSK & QPSK provided as standard; also supports 8PSK, 8QAM, 16QAM, 32APSK & 64QAM subject to selection of these options
	FastLink™ LDPC up to 1Mbps (requires FastLink LDPC hardware option)
	Extends FastLink™ LDPC to 2.5Mbps
	Extends FastLink™ LDPC to 5Mbps
	Extends FastLink™ LDPC to 10Mbps
	Extends FastLink™ LDPC to 25Mbps
	8QAM
	16APSK
8PSK (Includes TCM)	Note use of 8PSK other than with TCM requires either FastLink™ LDPC or TPC FEC option Rate 2/3 8PSK Pragmatic TCM to IESS 310
	16QAM (requires either FastLink™ LDPC or TPC FEC option)
Tx-only operation	Transmit functions only
Rx-only operation	Receive functions only
IBS	Satellite Framing to IESS 309 with low rate Intelsat ESC (to IESS 403) & High Rate IBS ESC
Drop / Insert (includes Extended D/I)	G.703 T1/E1 Drop & Insert; E1 CAS & T1 RBS signaling; Rx partial insert for multi-destinational working; timeslot ID maintenance for N=1 to 31
Clock Extension	Provides a high-stability reference clock over satellite (alternative to GPS)
Advanced AUX	Variable rate synchronous Aux channel; option to replace IDR audio channels with serial data
Custom	Custom Reed-Solomon values of n, k & interleaver depth; custom IBS modes; allocation of overhead between ESC & Aux; custom backward alarms
EZ BERT - PRBS Tester	Internal Bit Error Rate Tester
OM-73	OM-73 Scrambling, symbol mapping and Viterbi compatibility
48V DC Input	48V DC Primary power input in place of 100-240V AC input (hardware option)
FSK Control Option	Allows monitor & control of a compatible Transceiver from the Modem via the Tx IFL (hardware option)
Adaptive Signal Pre-distorter	Use with 16QAM to relax HPA backoff by up to 1.6dB. Compensates for HPA non-linearities in ground segment and/or transponder. Requires 16QAM option.
Ruggedisation	Adds extra ruggedisation for hostile environments (extra fans, heatsinks, etc.)
Sequential FEC	Rates 1/2, 3/4, 7/8 in BPSK, QPSK, OQPSK to 2.048Mbps
Audio	P1348 emulation mode for IBS 64kbps carrier (2 x audio) or 128kbps (2 x audio + 64kbps data) - requires IBS / SMS & IDR options

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